

# Companion to Bird & de Moor, Algebra of Programming

§5. Datatypes in Allegories

§5.1. Relators

(5.1a) A relator  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories: it preserves  $1$ , composition and  $\sqsubseteq$ .

|   |                                                                                                                                                                                                                                                                                              |        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|   | the statement                                                                                                                                                                                                                                                                                | (5.1b) |
| 1 | $\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$ . Lemma 5.1                                                                                                                                                                                   |        |
| 2 | Over a <b>tabular</b> allegory a functor is a relator $\iff$ it preserves $^\circ$ . Theorem 5.1 the book’s reverse direction (preserving $^\circ \implies$ relator) needs the source to be <b>Rel</b> : the counterexample over a tabular allegory that is not <b>Rel</b> shows it can fail |        |
| 3 | $F(R^\circ) = F(R)^\circ$ for every $R$ , so $F(R)^\circ$ needs no bracket.                                                                                                                                                                                                                  |        |
| 4 | Two relators agreeing on maps are equal. Corollary 5.1                                                                                                                                                                                                                                       |        |
| 5 | $\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$ Ex 5.2                                                                                                                                                                                            |        |
| 6 | $F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly. Ex 5.2                                                                                                                                                                                                                              |        |
| 7 | $F(\text{dom}(R)) = \text{dom}(F(R))$ Ex 5.5                                                                                                                                                                                                                                                 |        |

The **power relator**  $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \ \exists b \in ys. \ a \ R \ b) \wedge (\forall b \in ys. \ \exists a \in xs. \ a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

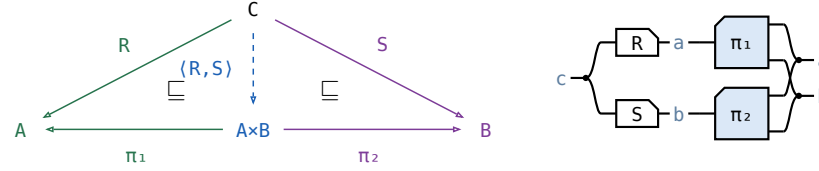
## §5.2. Relational products

$$\langle R, S \rangle \triangleq (R \pi_1^\circ) \cap (S \pi_2^\circ) \text{ — the fork of } R : C \rightarrow A \text{ and } S : C \rightarrow B, \text{ with } (\pi_1, \pi_2) \text{ the tabulation of } \tau. \quad (5.1)$$

$$\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ \quad c \in \langle R, S \rangle \iff c R a \wedge c S b. \quad (5.2a)$$

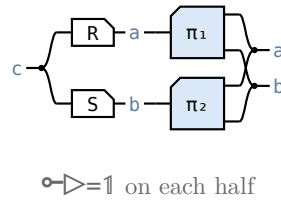
$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad (5.6)$$

$$\langle R, S \rangle \pi_2 = \text{dom}(R) S \quad (5.7)$$



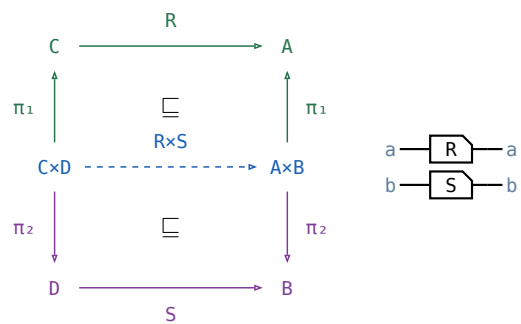
A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\mathbf{Rel}$ ,  $c \in \langle R, S \rangle \iff c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft(R \otimes S)$  on the right.

No  $^\circ$  survives the translation.  $\pi_1 = 1 \otimes \bullet$  discards the second component, so  $\pi_1^\circ = 1 \otimes \bullet$  **creates** one out of nothing, and  $\cap$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



### §5.2.1. Relational product $R \times S$

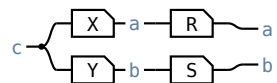
$$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle \text{ — the relational product } R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle. \quad (5.2)$$



Right-then-up is  $(R \times S) \pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S) \pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) \in (R \times S) \iff c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §5.2.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



(5.2.2a)

|   | the law                                                                                                                                                                                   | picture |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 | $\langle X, Y \rangle \langle R, S \rangle^\circ = (X \ R^\circ) \cap (Y \ S^\circ)$                                                                                                      |         |
| 2 | $\langle R, S \rangle^\circ \langle U, V \rangle \subseteq (R^\circ \ U) \times (S^\circ \ V)$                                                                                            |         |
| 3 | $\text{Map}(f) \Rightarrow f \langle R, S \rangle = \langle f \ R, f \ S \rangle$<br>f a map; it fails for an arbitrary arrow;                                                            |         |
| 4 | $X \langle R, S \rangle \subseteq \langle X \ R, X \ S \rangle$<br>the half of the row above that holds for every X                                                                       |         |
| 5 | $F(R \times S) \text{ unzip}(F) \subseteq \text{unzip}(F) (F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq \langle F(\pi_1), F(\pi_2) \rangle$ ; only $\subseteq$ for an arbitrary R, S |         |
| 6 | $g = \text{curry}(f) \iff (g \times 1) \text{ eval} = f$<br>reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?                                            |         |

## §5.3. Relational coproducts

(5.3a)

| the statement                                                                                                                                                                                                | picture |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R : a_1 \rightarrow c, S : a_2 \rightarrow c$                                                                                                                                                               |         |
| 1 $[R, S] \triangleq (l^\circ R) \cup (r^\circ S)$<br>(5.9) The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |
| 2 $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$                                                                                                                                                  |         |
| 3 $l [R, S] = R, r [R, S] = S$<br>$l T = R \wedge r T = S \Rightarrow T = [R, S],$                                                                                                                           |         |
| 4 $U : a_1 \rightarrow D, V : a_2 \rightarrow D$<br>$[U, V]^\circ [R, S] = (U^\circ R) \cup (V^\circ S) \quad (5.11)$                                                                                        |         |
| 5 $Z : c \rightarrow D$<br>$[R, S] Z = [R Z, S Z]$                                                                                                                                                           |         |

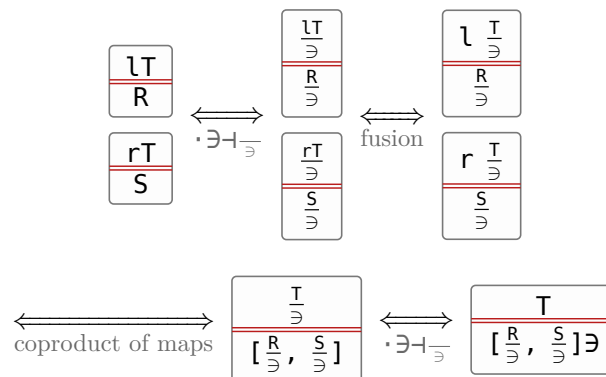
(5.3b)

| the statement                                             | picture                                                                                                                                                     |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $R : a_1 \rightarrow b_1, S : a_2 \rightarrow b_2$        |                                                                                                                                                             |
| 1 $R + S \triangleq [R l, S r] \quad (5.10)$              |                                                                                                                                                             |
| 2 $l l^\circ = 1, r r^\circ = 1 \quad \text{Ex 5.12}$     | $a_1 \xrightarrow{l} a \xrightarrow{l^\circ} a_1 = a_1 \xrightarrow{a} a_1 \quad a_2 \xrightarrow{r} a \xrightarrow{r^\circ} a_2 = a_2 \xrightarrow{a} a_2$ |
| 3 $l r^\circ = 0, r l^\circ = 0 \quad \text{Ex 5.12}$     | $a_1 \xrightarrow{l} a \xrightarrow{r^\circ} a_2 = a_1 \xrightarrow{0} a_2$                                                                                 |
| 4 $(l^\circ l) \cup (r^\circ r) = 1 \quad \text{Ex 5.12}$ |                                                                                                                                                             |

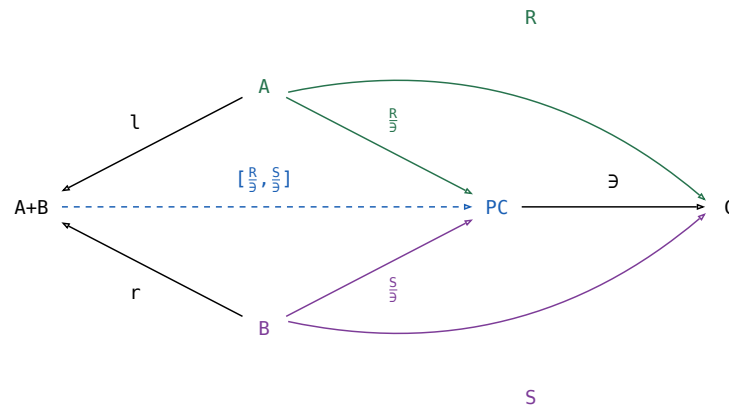
§5.3.1.  $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$  — a case of relations is the case of their transposes, then  $\exists$ 

The universal-property row is not free:  $l, r$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\exists$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow PC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,

(5.3.1a)



(5.3.1b)



Nothing here holds only up to  $\cong$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\bar{l}}, \frac{S}{\bar{r}}] \exists$ , and pushing  $\exists$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\bar{l} \circ \frac{R}{\bar{l}} \cup \bar{r} \circ \frac{S}{\bar{r}}) \exists = \bar{l} \circ R \cup \bar{r} \circ S$ .

(5.3.1c)

| the law                                                                                                                                                                                                                          | picture |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 prove (5.11), and say why duality does not carry it over from the product law                                                                                                                                                  |         |
| 2 $(R + S) \cap ([P, Q] [U, V]^\circ)$<br>$= (R \cap (P \cup^\circ)) + (S \cap (Q \cup^\circ))$<br>$[P, Q] [U, V]^\circ$ is a full $2 \times 2$ of composites; $R+S$ is diagonal, so the meet cuts the two off-diagonal branches |         |

§5.4. The power relator  $P(R)$ 

(5.4a)

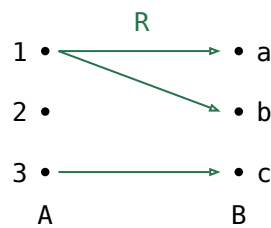
$P(R) \triangleq (\epsilon \setminus (R \epsilon)) \cap ((\exists R) / \exists)$  — the power relator  $P(R) : PA \rightarrow PB$ ,  $R$  lifted to sets.

$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$

(5.4b)

|   | in Rel                                | in words                                                                                                                                                           | im(xs)                       | partner                                       |
|---|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| 1 | $(\exists R) / \exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                                                                                    | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| 2 | $(\exists / (\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                                                                                    | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| 3 | $E(R)$                                | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                                                                                      | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| 4 | $((\exists R^\circ) / \exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                                                                                    | —                            | every $x$ has a partner                       |
| 5 | $P(R)$                                | $P(R)(xs, ys) \iff (\forall y. y \in ys \implies (\exists x. x \in xs \wedge R(x, y))) \wedge (\forall x. x \in xs \implies (\exists y. y \in ys \wedge R(x, y)))$ | —                            | every $x$ and every $y$ has a partner         |

(5.4c)



|   | xs            | $(\exists R) / \exists$ | $(\exists / (\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ) / \exists)^\circ$ | $P(R)$      |
|---|---------------|-------------------------|---------------------------------|-------------|---------------------------------------|-------------|
| 1 | $\emptyset$   | $\emptyset$             | all 8 subsets of abc            | $\emptyset$ | all 8 subsets of abc                  | $\emptyset$ |
| 2 | $\{1\}$       | $\emptyset, a, b, ab$   | ab, abc                         | ab          | a, b, ab, ac, bc, abc                 | a, b, ab    |
| 3 | $\{2\}$       | $\emptyset$             | all 8 subsets of abc            | $\emptyset$ | none                                  | none        |
| 4 | $\{3\}$       | $\emptyset, c$          | c, ac, bc, abc                  | c           | c, ac, bc, abc                        | c           |
| 5 | $\{1, 2\}$    | $\emptyset, a, b, ab$   | ab, abc                         | ab          | none                                  | none        |
| 6 | $\{1, 3\}$    | all 8 subsets of abc    | abc                             | abc         | ac, bc, abc                           | ac, bc, abc |
| 7 | $\{2, 3\}$    | $\emptyset, c$          | c, ac, bc, abc                  | c           | none                                  | none        |
| 8 | $\{1, 2, 3\}$ | all 8 subsets of abc    | abc                             | abc         | none                                  | none        |

(5.4d)

|   | law                                                                                                          | the reading                                                                                                                                                                                                                                                                                                    |
|---|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$ .                                                                                                                  |
| 2 | $P(1) = 1$                                                                                                   | The straightness axiom verbatim: extensionality <b>is</b> $P$ 's unit law.                                                                                                                                                                                                                                     |
| 3 | $\text{Map}(f) \implies P(f) = E(f)$                                                                         | In Rel, $xs \ P(f) \ ys \iff ys = \{f(a) \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f(a)$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| 4 | $P(R \ S) = P(R) \ P(S)$                                                                                     | $\exists$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                        |

§5.4.1. Type relators

(5.4.1a)

$T(R) = \langle F(R, 1) \ \alpha \rangle$  — the type functor  $T$  acts on an arrow  $R : A \multimap B$  by rebuilding with  $\alpha$ , applying  $R$  to the parameter.

(5.4.1b)

| name                    | law                                       | what it says                                                                                         |
|-------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1 the defining equation | $T(R) = \langle F(R, 1) \ \alpha \rangle$ | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                      |
| 2 type relator          | $T(R)^\circ = T(R^\circ)$                 | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map. |

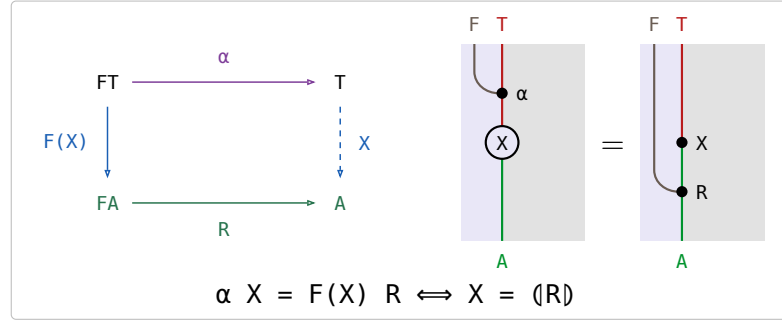
## §5.5. Relational catamorphisms

(5.5a)

The initial algebra  $\alpha : F(T) \rightarrow T$  among the maps is also initial in the allegory:

### §5.5.1. The defining equation

(5.5.1a)



An F-algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but F-Algebra only need it works on  $T$  and  $A$ .

(5.5.1b)

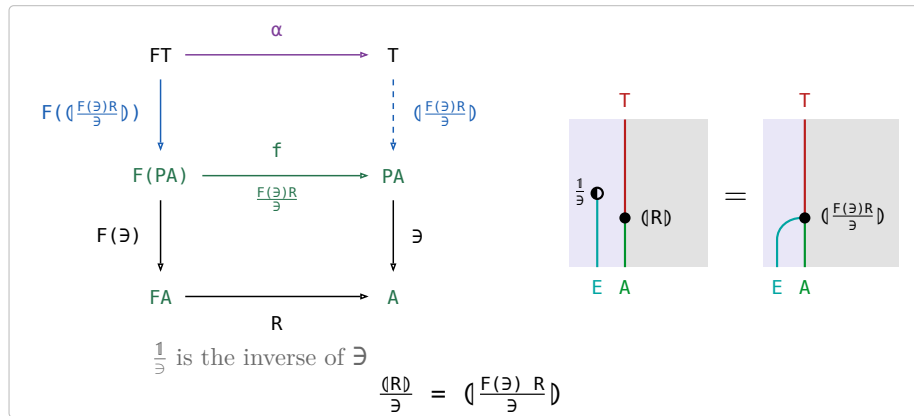
$$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$$

catamorphism: arrows of the form  $\llbracket f \rrbracket$  are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing

|   |                          |                                                                                                                                        |                                                                                                                                             |
|---|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>part</b>              | <b>Nat</b>                                                                                                                             | <b>[A]</b>                                                                                                                                  |
| 2 | datatype                 | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                                      | $[A] ::= \text{nil} \mid \text{cons}(A, [A])$                                                                                               |
| 3 | base functor F           | $F(X) = 1 + X$                                                                                                                         | $F(X) = 1 + A \times X$                                                                                                                     |
| 4 | initial algebra $\alpha$ | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                                     | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                                                |
| 5 | the fold, pointwise      | $\llbracket [c, f] \rrbracket(\text{zero}) = c$<br>$\llbracket [c, f] \rrbracket(\text{succ}(n)) = f(\llbracket [c, f] \rrbracket(n))$ | $\llbracket [c, f] \rrbracket(\text{nil}) = c$<br>$\llbracket [c, f] \rrbracket(\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket(x))$ |

### §5.5.2. $\llbracket R \rrbracket = \llbracket \frac{F(\exists) R}{\exists} \rrbracket \exists$ — folding with R is a fold of maps into sets, then $\exists$

(5.5.2a)



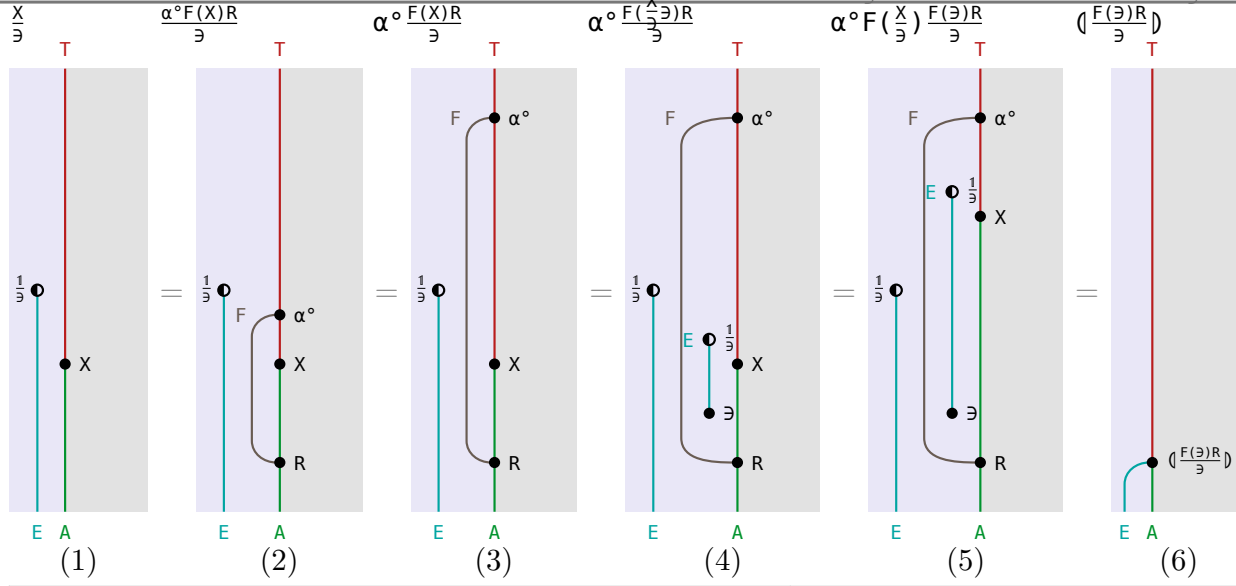
(5.12)

$$\begin{array}{c}
 \frac{\alpha X}{F(X)R} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}} \\
 \xleftrightarrow{\text{relator, fusion twice}} \frac{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}}{\frac{F(\frac{X}{\exists})}{\exists}} \xleftrightarrow{\text{reduce of maps}} \frac{\frac{X}{\exists}}{\frac{\llbracket \frac{F(\exists)R}{\exists} \rrbracket}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \frac{1}{\exists}} \frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}
 \end{array}$$

the last two rows at  $X := \llbracket R \rrbracket$ :

$$\alpha X = F(X) R \iff \frac{X}{\exists} = \langle \frac{F(\exists) R}{\exists} \rangle \quad (5.5.2b)$$

in a tabular allegory, with  $F$  a relator and  $\alpha : F(T) \rightarrow T$  its initial algebra: a relation  $X$  out of  $T$  satisfies the fold equation  $\alpha X = F(X) R$  of  $R$  exactly when  $\frac{X}{\exists}$  is the fold of the map  $\frac{F(\exists) R}{\exists}$



|   |                                                                    |                                                                                                                                                                                                                                                                          |
|---|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | $\frac{X}{\exists}$                                                | $\alpha$ iso; $\alpha X = F(X) R$                                                                                                                                                                                                                                        |
| 2 | $= \frac{\alpha^\circ F(X) R}{\exists}$                            | $\alpha^\circ$ is a map because $\alpha$ is an iso, so it leaves $\Lambda$ : $\frac{\alpha^\circ F(X) R}{\exists} = \alpha^\circ \frac{F(X) R}{\exists}$ — $\text{Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$                                       |
| 3 | $= \alpha^\circ \frac{F(X) R}{\exists}$                            | $X = \frac{X}{\exists} \frac{R}{\exists} \implies \frac{R}{\exists} \exists = R$                                                                                                                                                                                         |
| 4 | $= \alpha^\circ \frac{F(\frac{X}{\exists}) R}{\exists}$            | $F(\frac{X}{\exists}) = F(\frac{X}{\exists}) F(\exists)$ as $F$ is a functor, and $F(\frac{X}{\exists})$ is a map (a relator sends maps to maps), so it leaves $\Lambda$ — $\triangleq$ relator, $\text{Map}(f)$<br>$\implies \frac{f R}{\exists} = f \frac{R}{\exists}$ |
| 5 | $= \alpha^\circ F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$ | fold uniqueness, $\frac{F(\exists) R}{\exists}$ being a map — $(h : f \rightarrow g) \triangleq f h = F(h) g$ , $\text{Map}(f)$<br>$\implies \alpha \langle f \rangle = F(\langle f \rangle) f$                                                                          |
| 6 | $= \langle \frac{F(\exists) R}{\exists} \rangle$                   |                                                                                                                                                                                                                                                                          |

## §5.6. Combinatorial functions

(5.6a)

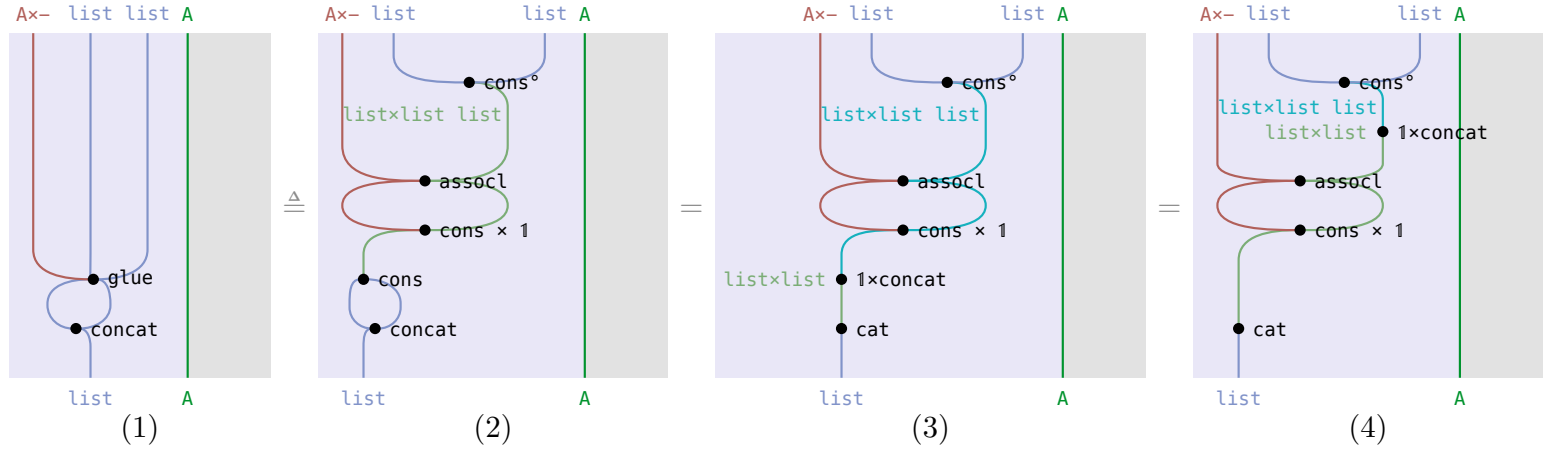
|    | definition                                                                                                                                                                                                                                                                            | type                                                                                                                            | note                                                                                                                                                                                                                                |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | $[A] ::= \text{nil} \mid \text{cons}(A, [A]) \quad \text{list}(A) = [A]$                                                                                                                                                                                                              | $\mathcal{A} \rightarrow \mathcal{A}$                                                                                           | The list type, under the short name it keeps.                                                                                                                                                                                       |
| 2  | $\text{list}(R) = \langle [\text{nil}, (R \times 1) \text{ cons}] \rangle$                                                                                                                                                                                                            | $[A] \rightarrow [B]$                                                                                                           | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                                                                                                           |
| 3  | $\text{subseq} = \langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$                                                                                                                                                                                                                | $[A] \rightarrow [A]$                                                                                                           | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — $\text{cons}$ keeps the head, $\pi_2$ drops it.                                                                                                                      |
| 4  | $\text{prefix} = \langle [\text{nil}, \text{cons} \cup \rightarrow \text{nil}] \rangle$<br>$\text{prefix} = \text{cat}^\circ \pi_1 = \text{init}^*$                                                                                                                                   | $[A] \rightarrow [A]$                                                                                                           | $ys$ is an initial segment of $xs$ ; the first $\text{nil}$ is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$ .                                                                                             |
| 5  | $\text{suffix} = \text{cat}^\circ \pi_2 = \text{tail}^*$                                                                                                                                                                                                                              | $[A] \rightarrow [A]$                                                                                                           | The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$ ; as a reduce it needs $\text{snoc}$ -lists.                                                                                                                             |
| 6  | $\text{partition} = \text{concat}^\circ$                                                                                                                                                                                                                                              | $[A] \rightarrow [[A]]$                                                                                                         | This $\text{cat}$ is restricted to $[A]^+ \times [A] \rightarrow [A]$ , so $ys$ is a list of non-empty segments of $xs$ .                                                                                                           |
| 7  | $\text{concat} = \langle [\text{nil}, \text{cat}] \rangle$                                                                                                                                                                                                                            | $[[A]] \rightarrow [A]$                                                                                                         | Joins the segments back up, which is why its converse splits a list.                                                                                                                                                                |
| 8  | $\text{inits}$                                                                                                                                                                                                                                                                        | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{prefix}}{\exists}$ , listing the prefixes by increasing length.                                                                                                                                             |
| 9  | $\text{tails}$                                                                                                                                                                                                                                                                        | $[A] \rightarrow [[A]]$                                                                                                         | Implements $\frac{\text{suffix}}{\exists}$ by decreasing length — the opposite order.                                                                                                                                               |
| 10 | $\text{cpp}, \text{cpl}, \text{cpr}$                                                                                                                                                                                                                                                  | $[A] \times [B] \rightarrow [A \times B]$<br>$[A] \times B \rightarrow [A \times B]$<br>$A \times [B] \rightarrow [A \times B]$ | Every pair $(a, b)$ with $a$ from the first list and $b$ from the second; $\text{cpl}$ , $\text{cpr}$ keep one side a single element.                                                                                               |
| 11 | $\text{cpp setify} = (\text{setify} \times \text{setify}) \frac{\exists \times \exists}{\exists}$<br>$\text{cpl setify} = (\text{setify} \times 1) \frac{\exists \times 1}{\exists}$<br>$\text{cpr setify} = (1 \times \text{setify}) \frac{1 \times \exists}{\exists}$               | $[A] \times [B] \rightarrow P(A \times B)$                                                                                      | Forgetting the order with $\text{setify}$ , $\text{cpp}$ is the transpose of $\exists \times \exists$ .                                                                                                                             |
| 12 | $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$                                                                                                                                                                                                                                  | $F(PA) \rightarrow PFA$                                                                                                         | $\text{cp}(F)$ , the transpose of a relator's action on $\exists$ ; $\text{cpp}$ implements it at $F(A) = A \times A$ .                                                                                                             |
| 13 | $\text{cp}(\text{list}) = \langle \frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} \rangle$<br>$\frac{[\text{nil}, (\exists \times \exists) \text{ cons}]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{\exists \times \exists}{\exists}]$<br>$E(\text{cons})]$ | $\text{list}(PA) \rightarrow P(\text{list}(A))$                                                                                 | $\text{cp}(\text{list})$ : (5.12) $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$ at $\text{list}(\exists) = \langle [\text{nil}, (\exists \times 1) \text{ cons}] \rangle$ , then the algebra expanded. |
| 14 | $\text{cplist} = \langle [\text{nil} \text{ wrap}, \text{cpp list}(\text{cons})] \rangle$                                                                                                                                                                                             | $[[A]] \rightarrow [[A]]$                                                                                                       | $\text{cp}(\text{list})$ with every set a list: $\text{cpp}$ in place of $\frac{\exists \times \exists}{\exists}$ .                                                                                                                 |

**glue concat  $\sqsubseteq (1 \times \text{concat}) \text{ cons}$** 

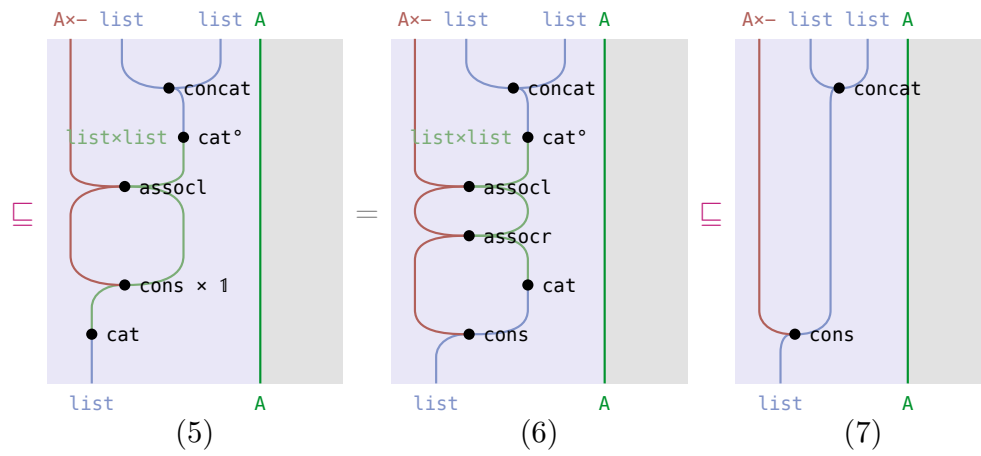
(5.6b)

putting **a** on the front of the first segment and flattening (**glue concat**) gives at most flattening the segments and putting **a** on the front of the result (**(1×concat) cons**)

the starting composite



|   |                                                                                                              |                                                    |
|---|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 1 | <b>glue concat</b>                                                                                           | definition of glue                                 |
| 2 | $\triangleq (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cons } \text{concat}$   | since $\text{concat} = ([\text{nil}, \text{cat}])$ |
| 3 | $= (1 \times \text{cons}^\circ) \text{ assocl } \text{cons} \times 1 (1 \times \text{concat}) \text{ cat}$   | naturality of $\text{assocl}$                      |
| 4 | $= (1 \times (\text{cons}^\circ (1 \times \text{concat}))) \text{ assocl } \text{cons} \times 1 \text{ cat}$ | since $\text{concat} = ([\text{nil}, \text{cat}])$ |



|   |                                                                                                                            |                                                                                                |
|---|----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 5 | $\sqsubseteq (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{cons} \times 1 \text{ cat}$        | since $(\text{cons} \times 1) \text{ cat} = \text{assocr } (1 \times \text{cat}) \text{ cons}$ |
| 6 | $= (1 \times \text{concat}) (1 \times \text{cat}^\circ) \text{ assocl } \text{assocr } (1 \times \text{cat}) \text{ cons}$ | since $\text{assocl } \text{assocr} = 1$ and $\text{cat}^\circ \text{ cat} \sqsubseteq 1$      |
| 7 | $\sqsubseteq (1 \times \text{concat}) \text{ cons}$                                                                        |                                                                                                |

**§5.6.1.**  $\frac{\text{subseq}}{\exists} = ([\text{nil } \frac{1}{\exists}, (\frac{1 \times \exists}{\exists} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$  — the set of subsequences is a fold of maps into sets

(5.6.1a)

$\text{cup} \triangleq \frac{(\pi_1 \exists) \cup (\pi_2 \exists)}{\exists}$  — cup unions two sets.

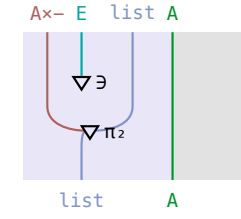
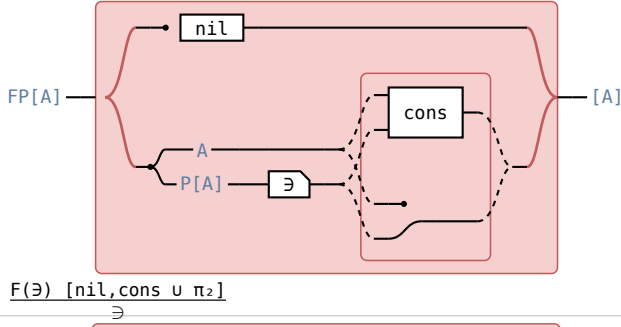
$\text{cup} \triangleq \frac{\pi_1 \exists \cup \pi_2 \exists}{\exists} : \text{PA} \times \text{PA} \rightarrow \text{PA}$

$$\frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}] \quad (5.6.1b)$$

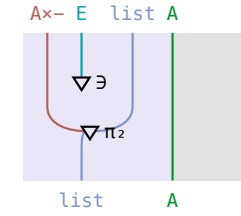
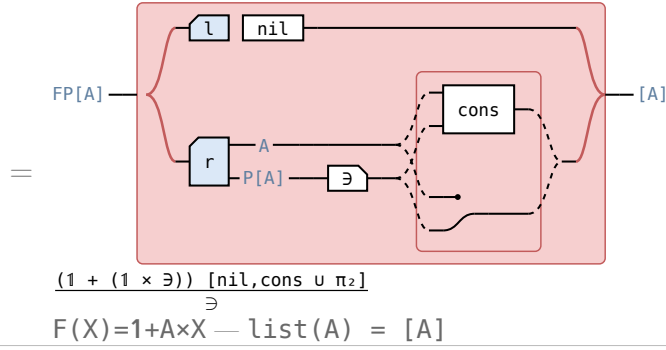
the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — (5.12)  $\langle R \rangle = \langle \frac{F(\exists)}{\exists} R \rangle \exists$  at **subseq** =  $\langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$ , **subseq** =  $\langle [\text{nil}, \text{cons} \cup \pi_2] \rangle$ .

**circuit** — the fork is  $F([A])=1+A \times [A]: \text{nil}$  above, the pair below

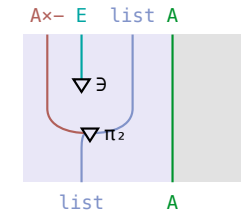
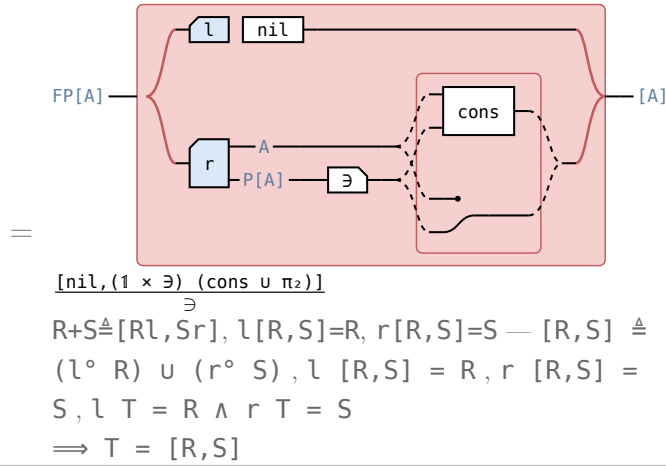
**Hinze–Marsden**



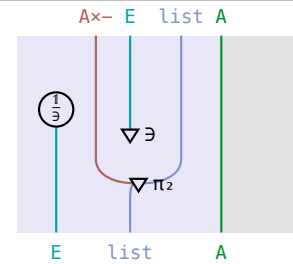
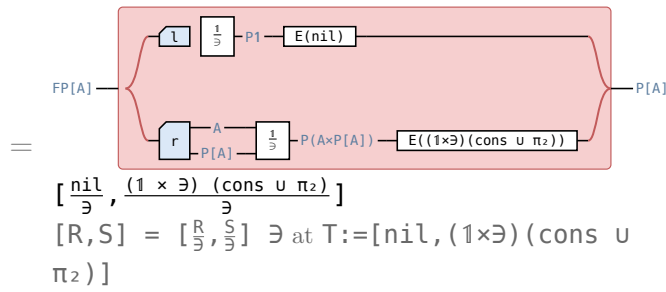
the  $\pi_2$  operand of  $\text{cons} \cup \pi_2$  under the  $1 \times \exists$  summand of  $F(\exists)$ , i.e.  $(1 \times \exists) \pi_2$



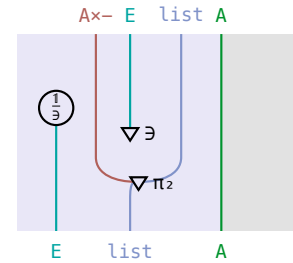
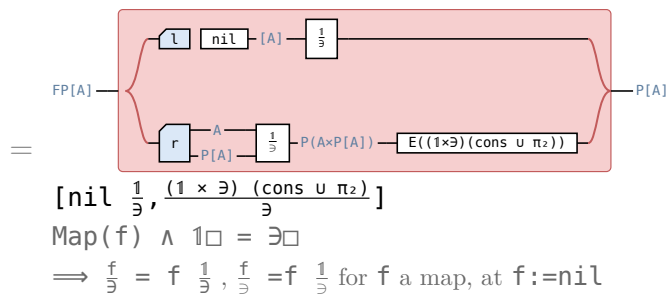
the same operand under  $1+1 \times \exists$ , whose  $1 \times \exists$  summand it sits in



the  $\pi_2$  operand of the second arm  $(1 \times \exists) (\text{cons} \cup \pi_2)$

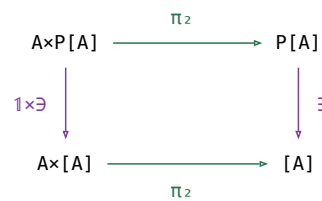


the  $\pi_2$  operand under its  $\frac{1}{\exists}$ , the arm  $(1 \times \exists) \pi_2 = \pi_2 \exists$  rewrites



the same operand; the two rows differ only in the **nil** arm

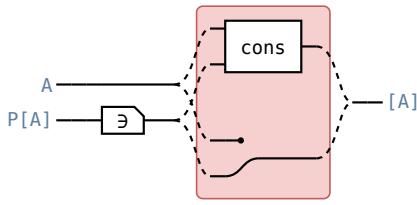
(5.6.1c)



$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \langle \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \rangle \text{cup} \quad (5.6.1d)$$

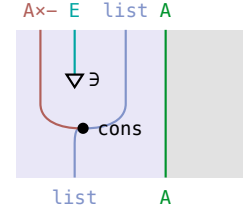
power transpose of join: the power transpose of the join of two relations is  $\langle \frac{R}{\exists}, \frac{S}{\exists} \rangle \text{cup}$ , where **cup** is the function that returns the union of two sets

**circuit**

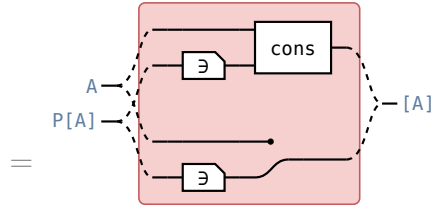


$$\frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists} = \frac{F(\exists) [\text{nil}, \text{cons} \cup \pi_2]}{\exists} = [\text{nil} \frac{1}{\exists}, \frac{(1 \times \exists) (\text{cons} \cup \pi_2)}{\exists}] \text{'s second branch}$$

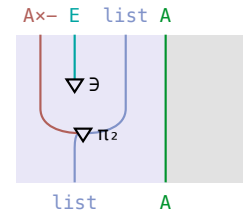
**Hinze–Marsden**



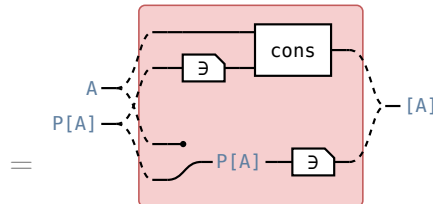
the **cons** operand of  $\text{cons} \cup \pi_2$



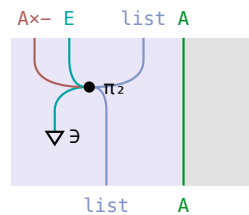
$$\frac{((1 \times \exists) \text{cons}) \cup ((1 \times \exists) \pi_2)}{\exists} = \frac{T(X_1 \cup X_2) = TX_1 \cup TX_2 \text{ — } (S \cup T) R = (S R) \cup (T R), R(S \cup T) = (R S) \cup (R T), R \cap (S \cup T) = R \cap S \cup R \cap T, (R \cup S)^\circ = S^\circ \cup R^\circ}{\exists}$$



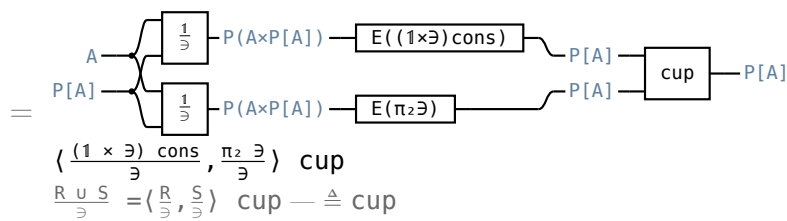
the  $\pi_2$  operand of  $(1 \times \exists) \text{cons} \cup (1 \times \exists) \pi_2$



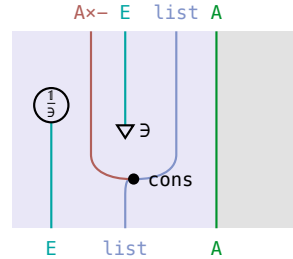
$$\frac{((1 \times \exists) \text{cons}) \cup ((\pi_2 \exists))}{\exists} = \frac{(1 \times \exists) \pi_2 = \pi_2 \exists \text{ — } (R \times S) \pi_1 \sqsubseteq \pi_1 R \wedge (R \times S) \pi_2 \sqsubseteq \pi_2 S \text{ at } \pi_2, \text{ an equality because } 1 \text{ is entire}}{\exists}$$



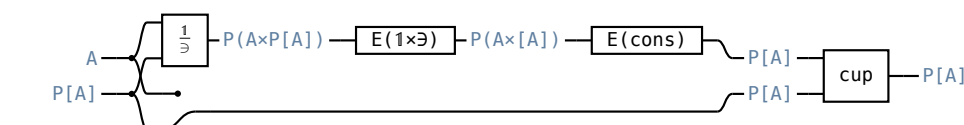
the  $\pi_2$  operand of  $(1 \times \exists) \text{cons} \cup \pi_2 \exists$



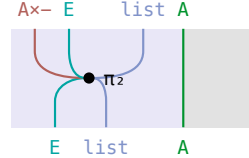
$$\frac{\langle \frac{1 \times \exists}{\exists} \text{cons}, \frac{\pi_2 \exists}{\exists} \rangle \text{cup}}{\frac{R \cup S}{\exists} = \langle \frac{R}{\exists}, \frac{S}{\exists} \rangle \text{cup} \text{ — } \triangleq \text{cup}}$$



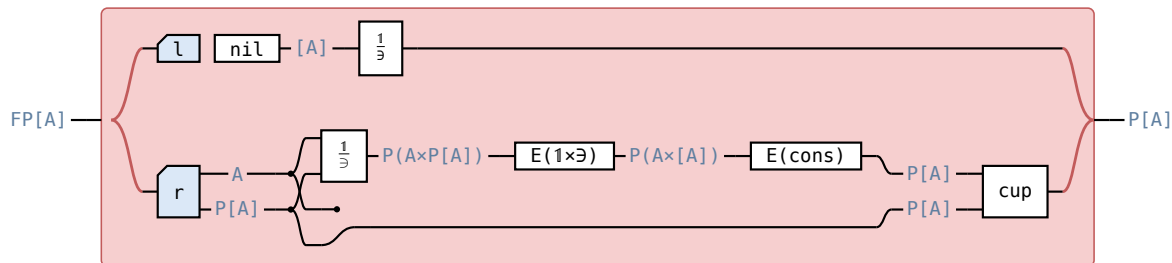
the **cons** operand under its  $\frac{1}{\exists}$



$$\frac{\langle \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \rangle \text{cup}}{\frac{S}{\exists} E(R) = \frac{S}{\exists} R, \text{ absorption } \frac{S}{\exists} E(R) = \frac{SR}{\exists} \text{ at } S := 1 \times \exists, R := \text{cons}; \text{ fusion and } \frac{\exists}{\exists} = 1 \text{ on the } \pi_2 \text{ operand}}$$



the  $\pi_2$  operand, bare  $\pi_2$



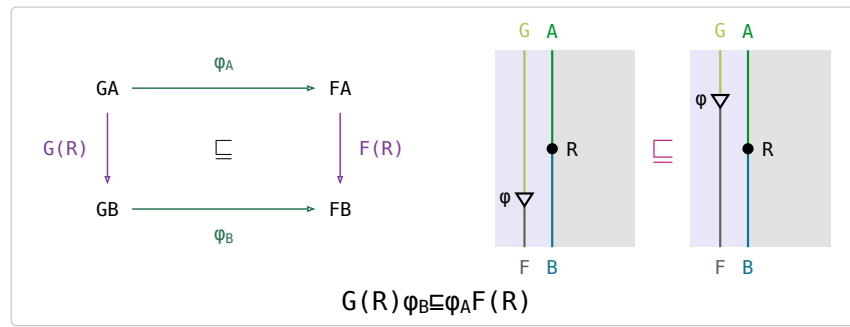
$$[\frac{\text{nil}}{\exists}, \langle \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \rangle \text{cup}]$$

which writes  $P\text{cons}$ ; **cons** is a map, and there  $P(\text{cons}) = E(\text{cons}) \text{ — Map}(f) \Rightarrow P(f) = E(f)$ .

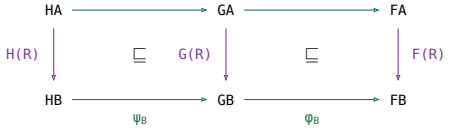
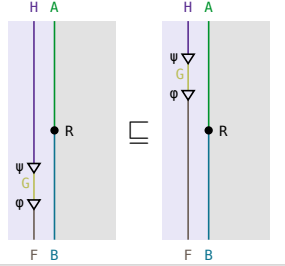
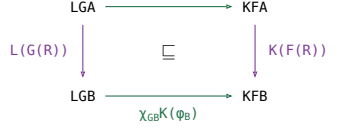
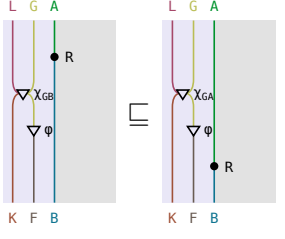
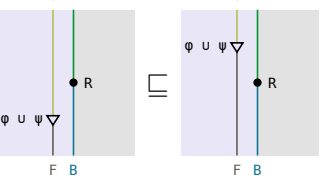
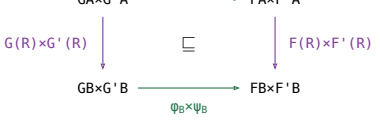
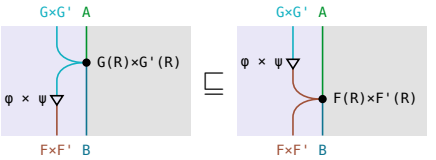
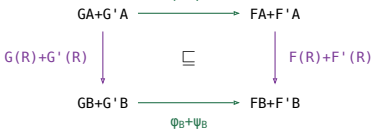
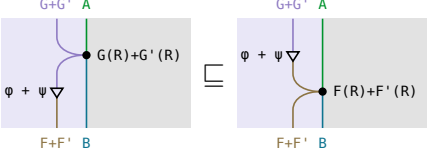
(5.6.1e)

## §5.7. Lax natural transformations (LaT)

$\varphi : G \Rightarrow F$  is **lax** at  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$ ; it is a **lax natural transformation** (LaT) when lax at every  $R$ . (5.7a)



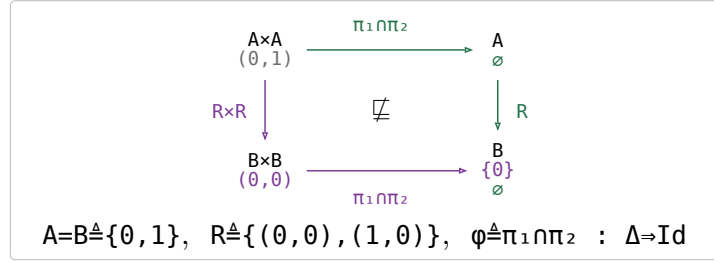
(5.7c)

|   | closed under                                                                                                                                                                                                                                                                                                                                                                                                                                                               | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | string diagram                                                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 | <p>composition<br/><math>\psi\phi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                               |  <p><math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    |
| 2 | <p>horizontal composition<br/><math>\chi\circ\phi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                               |  <p><math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi\circ\phi : L\circ G \Rightarrow K\circ F</math>, the family <math>A \mapsto \chi_{GA}K(\phi_A)</math><br/> <math>L(G(R))\chi_{GB} \sqsubseteq \chi_{GA}K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation;<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K\circ G \Rightarrow K\circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math> gives <math>\chi G : L\circ G \Rightarrow K\circ G</math>, <math>G</math> composed on the inside<br/> two candidates, <math>\chi_{GA}K(\phi_A)</math> and <math>L(\phi_A)\chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is the first</p> |    |
| 3 | <p>union<br/><math>\phi \cup \psi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                               |  <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A)F(R)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| 4 | <p>a relator <math>K</math><br/><math>K(\phi)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                   |  <p><math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R))K(\phi_B) \sqsubseteq K(\phi_A)K(F(R))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                       |
| 5 | <p>product<br/><math>\phi \times \psi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                           |  <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S)(U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork is the derived case <math>\langle \phi, \psi \rangle = \triangleleft(\phi \times \psi)</math>, and its ONE cost is the lax copy law <math>R \triangleleft \sqsubseteq \triangleleft(R \times R)</math> — <math>R \triangleleft \sqsubseteq \triangleleft(R \otimes R)</math>, <math>R \multimap \sqsubseteq \multimap</math></p>                                                                                                                                                                                                                                                                                                                                    |  |
| 6 | <p>coproduct<br/><math>\phi + \psi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                              |  <p><math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S)(U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \nabla</math>, and <math>\nabla</math> costs nothing<br/> the step it would need is <math>(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)</math>, the wrong direction of <math>R</math> (<math>S \cap T \sqsubseteq (R S) \cap (R T)</math>)<br/> a counterexample: <math>aa = (\emptyset, 1) \wedge bb = (\emptyset, 0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{\emptyset\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1)</math><br/> <math>\Rightarrow \neg((R \times R)(\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)</math></p>                                                 |  |
| 7 | <p>meet — fails<br/><math>\phi \cap \psi</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                       |
| 8 | <p>Each row is the general slide <math>R_A X \sqsubseteq X' R_B</math> and <math>R_B Y \sqsubseteq Y' R_C</math> give <math>R_A(XY) \sqsubseteq (X'Y')R_C</math> at <math>R_A, R_B := G(R), F(R)</math> and <math>X, X' := \phi_B, \phi_A</math>, quantified over <math>R</math>; at <math>X' = X</math> with <math>R_A, R_B</math> endorelations it is the monotonicity reading instead — the <math>\forall R</math> sits outside the law, and is the only difference</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                       |

## §5.7.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathcal{G} \Rightarrow \mathcal{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

(5.7.1a)



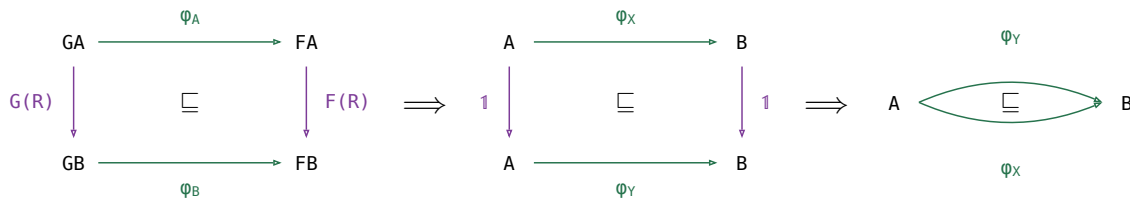
## §5.7.2. Two stacked towers

(5.7.2a)

|                | lower — inside ONE allegory      | upper — between allegories                                                                                                                                                                                                                                                                                                                                   |
|----------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 0-cell       | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                                                                                                                                                                                                                                                                              |
| 2 1-cell       | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                                                                                                                                                                                                                                                                                          |
| 3 2-cell       | $R \sqsubseteq S$                | a LaT $\varphi : \mathcal{G} \Rightarrow \mathcal{F} \triangleq \text{lax}$                                                                                                                                                                                                                                                                                  |
| 4 order        | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                                                                                                                                                                                                                                                                                     |
| 5 union        | $R \cup S$                       | $\varphi \cup \psi$ , survives $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow G(R) (\varphi_B \cup \psi_B) \sqsubseteq (\varphi_A \cup \psi_A) F(R)$                                                                                                                                                                                               |
| 6 product      | $R \times S \triangleq \times$   | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC $\langle R, S \rangle$<br>$\pi_1 = \text{dom}(S) R, \langle R, S \rangle \pi_2 = \text{dom}(R) S$ $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) \times G'(R)) (\varphi_B \times \psi_B) \sqsubseteq (\varphi_A \times \psi_A) (F(R) \times F'(R))$         |
| 7 coproduct    | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT $\text{lax}(\varphi) \wedge \text{lax}(\psi) \Rightarrow (G(R) + G'(R)) (\varphi_B + \psi_B) \sqsubseteq (\varphi_A + \psi_A) (F(R) + F'(R))$                                                                                                                                                                       |
| 8 meet         | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails $aa = (0,1) \wedge bb = (0,0) \wedge (R \times R)(aa, bb) \wedge b_1 = \{0\} \wedge b_1 = E(\pi_1 \cap \pi_2)\{bb\} \wedge a_1 = \emptyset \wedge a_1 = E(\pi_1 \cap \pi_2)\{aa\} \wedge b_2 = \emptyset \wedge b_2 = E(R)(a_1) \Rightarrow \neg((R \times R) (\pi_1 \cap \pi_2) \sqsubseteq (\pi_1 \cap \pi_2) R)$ |
| 9 converse     | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ G(R) \sqsubseteq F(R) \varphi_B^\circ$                                                                                                                                                                                                                                                                      |
| 10 zero object | $z$ with $1 \ z = \emptyset$     | the constant relator at such a $z$ , initial and terminal                                                                                                                                                                                                                                                                                                    |

## §5.7.3. RES is a special case of LaT

(5.7.3a)

 $G \triangleq \text{const } A, F \triangleq \text{const } B$  — the relators  $X \rightarrow A$  and  $X \rightarrow B$ , each sending every arrow to  $1 \triangleq \text{lax}$ with  $\mathcal{C}$  two objects and one non-identity arrow  $X \rightarrow Y$ , a LaT  $\text{const } A \Rightarrow \text{const } B$  is the pair  $\varphi_X, \varphi_Y : A \rightarrow B$  with  $\varphi_Y \sqsubseteq \varphi_X$  — the datum  $R \sqsubseteq S$ not every LaT is constant:  $\exists : \mathcal{P} \Rightarrow \text{Id} \ X \sqsubseteq \mathcal{P}(R) \iff X \exists \sqsubseteq \exists R \wedge X^\circ \exists \sqsubseteq \exists R^\circ \ \pi_2 : (A \times -) \Rightarrow \text{Id} \ (1 \times \exists) \pi_2 = \pi_2 \exists$  $\triangleleft : \text{Id} \Rightarrow \Delta \ R \triangleleft \leq \triangleleft (R \otimes R), R \multimap \leq \multimap$